

Exercise

(01)

```
1 #include <stdio.h>
2
3 int main() {
4     double weight_A = 68.0;
5     double weight_B = 75.0;
6     double weight_C = 50.0;
7
8     double height_A = 1.65;
9     double height_B = 1.70;
10    double height_C = 1.55;
11
12    double BMI_A = weight_A / (height_A * height_A);
13    double BMI_B = weight_B / (height_B * height_B);
14    double BMI_C = weight_C / (height_C * height_C);
15
16    double avg_BMI = (BMI_A + BMI_B + BMI_C) / 3.0;
17
18    double max_BMI, min_BMI;
19
20    max_BMI = BMI_A * (BMI_A > BMI_B) + BMI_B * (1 - (BMI_A > BMI_B));
21    max_BMI = max_BMI * (max_BMI > BMI_C) + BMI_C * (1 - (max_BMI > BMI_C));
22
23    min_BMI = BMI_A * (BMI_A < BMI_B) + BMI_B * (1 - (BMI_A < BMI_B));
24    min_BMI = min_BMI * (min_BMI < BMI_C) + BMI_C * (1 - (min_BMI < BMI_C));
25
26    printf("Average BMI of A, B and C    : %.2f kgm^-2\n\n", avg_BMI);
27
28    printf("Person A : \n\tBMI : %.2f kgm^-2\n", BMI_A);
29    if (BMI_A > avg_BMI) {
30        printf("\tYour BMI is higher than the average BMI\n");
31    } else {
32        printf("\tYour BMI is lower than the average BMI\n");
33    }
34    if (BMI_A < 24.9 && BMI_A > 18.5) {
35        printf("\tYour BMI is in the normal range (18.5 , 24.9)\n\n");
36    } else {
37        printf("\tYour BMI is outside the normal range (18.5, 24.9)\n\n");
38    }
```

```

39
40 printf("Person B : \n\tBMI : %.2f kgm^-2\n", BMI_B);
41 if (BMI_B > avg_BMI) {
42     printf("\tYour BMI is higher than the average BMI\n");
43 } else {
44     printf("\tYour BMI is lower than the average BMI\n");
45 }
46 if (BMI_B < 24.9 && BMI_B > 18.5) {
47     printf("\tYour BMI is in the normal range (18.5 , 24.9)\n\n");
48 } else {
49     printf("\tYour BMI is outside the normal range (18.5, 24.9)\n\n");
50 }
51
52 printf("Person C : \n\tBMI : %.2f kgm^-2\n", BMI_C);
53 if (BMI_C > avg_BMI) {
54     printf("\tYour BMI is higher than the average BMI\n");
55 } else {
56     printf("\tYour BMI is lower than the average BMI\n");
57 }
58 if (BMI_C < 24.9 && BMI_C > 18.5) {
59     printf("\tYour BMI is in the normal range (18.5 , 24.9)\n\n");
60 } else {
61     printf("\tYour BMI is outside the normal range (18.5, 24.9)\n\n");
62 }
63
64 printf("Maximum BMI from A, B and C : %.2f kgm^-2\n", max_BMI);
65 printf("Minimum BMI from A, B and C : %.2f kgm^-2\n", min_BMI);
66
67 return 0;
68 }

```

Average BMI of A, B and C : 23.91 kgm⁻²

Person A :

BMI : 24.98 kgm⁻²

Your BMI is higher than the average BMI

Your BMI is outside the normal range (18.5, 24.9)

Person B :

BMI : 25.95 kgm⁻²

Your BMI is higher than the average BMI

Your BMI is outside the normal range (18.5, 24.9)

Person C :

BMI : 20.81 kgm⁻²

Your BMI is lower than the average BMI

Your BMI is in the normal range (18.5 , 24.9)

Maximum BMI from A, B and C : 25.95 kgm⁻²

Minimum BMI from A, B and C : 20.81 kgm⁻²

(02)

```
1 #include <stdio.h>
2 #include <math.h>
3
4 int main() {
5     // rectangle dimensions
6     double length = 10;
7     double breadth = 5;
8
9     double area_rect = length * breadth;
10    double perimeter_rect = (length + breadth) * 2.0;
11    double diagonal = sqrt((length * length) + (breadth * breadth));
12
13    // triangle dimensions
14    double base = 8;
15    double height = 6;
16    double side_1 = 8;
17    double side_2 = 6;
18    double side_3 = 7;
19
20    double area_tri = 0.5 * base * height;
21    double perimeter_tri = side_1 + side_2 + side_3;
22
23    // circle dimensions
24    double radius = 7;
25    double area_cir = M_PI * radius * radius;
26    double circumference = 2.0 * M_PI * radius;
27    double diameter = 2.0 * radius;
28    double semi_area = M_PI_2 * radius * radius;
29
30    printf("Rectangle : \n\tArea      : %.2f m^2\n\tPerimeter : %.2f m\n\tDiagonal : %.2f m\n\n", area_rect, perimeter_rect, diagonal);
31    printf("Triangle : \n\tArea      : %.2f m^2\n\tPerimeter : %.2f m\n\n", area_tri, perimeter_tri);
32    printf("Circle : \n\tArea      : %.2f m^2\n\tCurcumference : %.2f m\n\tDiameter : %.2f m\n\tArea of Semicircle : %.2f m^2\n\n", area_cir, circumf
33    erence, diameter, semi_area);
34
35    return 0;
36 }
```

Rectangle :

Area : 50.00 m²

Perimeter : 30.00 m

Diagonal : 11.18 m

Triangle :

Area : 24.00 m²

Perimeter : 21.00 m

Circle :

Area : 153.94 m²

Curcumference : 43.98 m

Diameter : 14.00 m

Area of Semicircle : 76.97 m²